

Exploratory Data Analysis Summary: Titanic Dataset

This summary presents key insights derived from statistical and visual analysis of the Titanic dataset.

1. Dataset Overview:

- Contains demographic and survival information of passengers.
- Includes features like age, sex, fare, class, and survival.

2. Statistical Insights:

- Survival Rate: Approximately 38% survived.
- Gender Impact: Females had a higher survival rate than males.
- Class Difference: 1st class passengers had better chances of survival.

3. Visual Insights:

- Heatmap reveals strong correlation between class, sex, and survival.
- Histograms show most passengers were in 3rd class and between ages 20-40.
- Boxplots highlight fare distribution among classes.

4. Conclusions:

- Survival was significantly influenced by passenger class and gender.
- Data reveals socio-economic factors impacted survival rates.

This analysis helps understand passenger dynamics and critical survival indicators from the Titanic tragedy.

Tools Used: Python, Pandas, Matplotlib, Seaborn